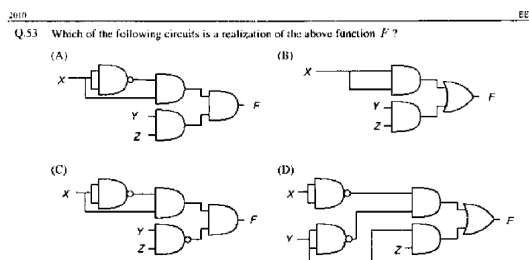


GATE Question Paper 2010, EE Question Number 53

Abstract



(GATE 2010, Question No. 53 – Implementing a Boolean logic function using Arduino)

1. Components

Component	Qty
Arduino UNO Board	1
USB Cable (Type B)	1
Push Buttons	3
LEDs	1
220Ω Resistors	3
Jumper Wires (M-M)	10
Breadboard	1
Android Mobile with Arduino App	1

Table 1: List of components used

2. Setup and Connections

1. Connect push buttons to D2, D3, D4 for X, Y, Z.
2. Add pull-down resistors.
3. Connect LED to pin D13 via 220Ω resistor.

4. Common ground to buttons and LED.

5. Power Arduino via USB and Arduino app.

3. Steps for Implementation

1. Complete the circuit.
2. Connect Arduino to mobile via USB.
3. Open Arduinodroid, select board and port.
4. Open, save, compile and upload code.

4. Truth Table

X	Y	Z	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

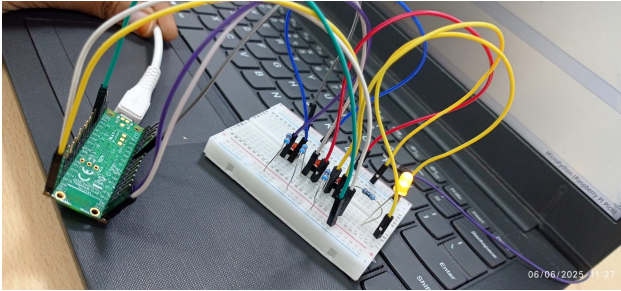
5. Boolean Expression Simplification

$$F = \overline{X}\overline{Y}\overline{Z} + \overline{X}\overline{Y}Z + \overline{X}YZ + XYZ$$

$$F = \overline{X}\overline{Y}(\overline{Z}+Z)+YZ(\overline{X}+X) \Rightarrow \overline{X}\overline{Y}+YZ$$

6. Input and Output Pins

- X (Input) – D2
- Y (Input) – D3
- Z (Input) – D4



7. Arduino Code Link

https://github.com/madhuu34/Madhulathafwc/blob/main/Hardware/PLATFORM_IO/hardware_code.py